Case Study Report

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Introduction

In this case study, I looked at simulations of tiny particles, such as water droplets in clouds for example, moving through turbulent air. So understanding how these particles come together to form clusters is very important as it can help explain things like how rain forms. The data set that I worked on contains 89 simulation runs in which each run involves three input variables. These variables include the Reynolds number (Re) which measures how turbulent the flow is, the Froude number (Fr) which tells how strong gravity is, and the Stokes number (St) which measures how much inertia the particles have. And for each simulation, I analyzed the distribution of cluster volumes by looking at four statistical measures, which includes mean, variance, skewness, and kurtosis. From my understanding, the mean tell us the average cluster volume, the variance measures how much the cluster volumes differ from the average, skewness describes whether the distribution is more spread out toward smaller or larger clusters, and kurtosis helps us understand how likely the distribution is to contain unusually small or large clusters.

With this case study, I got two goals in mind. My first goal is that I want to predict these four statistical measures for new set of combinations of Re, Fr, and St. My second goal is that I want to understand how each input variable affects the cluster volume distribution and whether these effects differ across the four statistical measures. To do this, I use linear regression models on the log scale, which allows for a nonlinear relationship with St and an interaction between Re and gravity. I then select my models using cross-validation to evaluate how well these models predict new data. With this approach, it allows me to understand the influence of each variable while also keeping the models relatively simple and accurate.

Methodology

Preparing the data

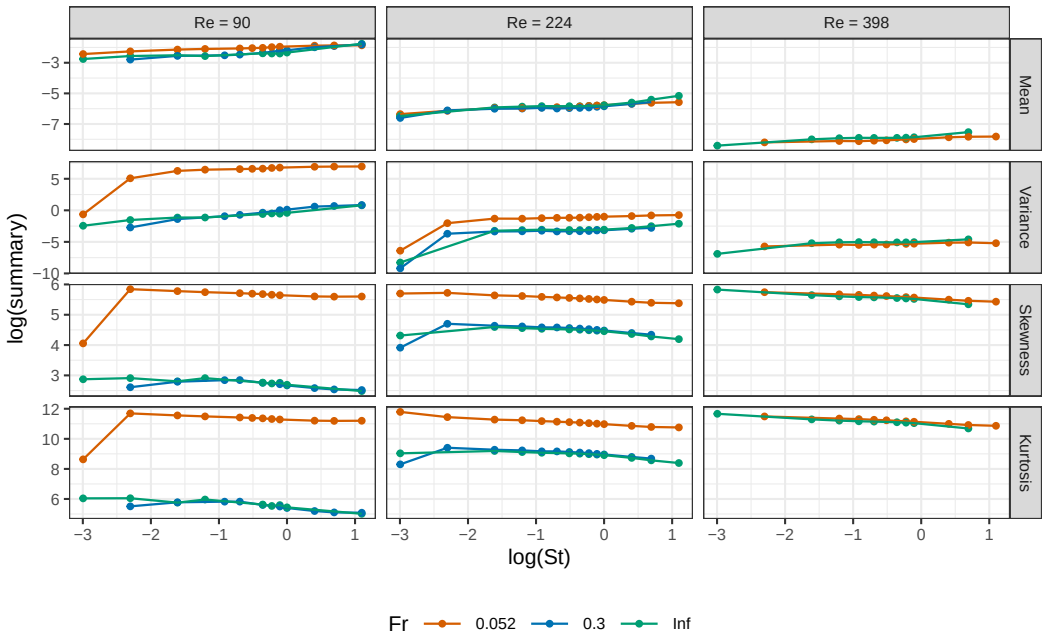
The simulations that were first provided to me has the first four raw moments of the cluster volumes, in which I use to calculate the mean, variance, skewness, and kurtosis using standard statistical formulas such as $E[X^2] - (E[X]^2)$ to calculate the variance as an example.

I found that all four statistical measures do have a wide range of values. Like for the variance, it ranges from about 10^{-4} to 10^3, which means that some values are much larger than others. The variation also varies significantly between different simulation settings. So to account for these differences, I applied a log transformation to each measure before building my models. With this, it makes the data easier

to analyze and it helps my models be better at understanding how each input variable influences the results.

For the input variables, I noticed that Re and Fr each have only three possible values. Thus I treated them as categorical variables. St has 14 different values in which most of them, bunched near 0, so I used log(St) to spread out these values and make it easier for me to understand how St affects the cluster volumes.

What I saw in the data



This figure has gave me some clear insights.

The first thing that I noticed is that the relationship with St is not linear. From the graphs, even after I applied a log transformation to St, the relationship is not a straight line. Because of this, I tried to include polynomial terms in our models to better capture these nonlinear patterns.

The second thing that I see is that Re and Fr interact. When gravity is strong like Fr = 0.052, I see that the variance, skewness, and kurtosis have all increased significantly at Re = 90. But I do see that this increase becomes smaller at Re = 224 and is barely noticeable at Re = 398. What this could mean is that the effect of gravity does depends on the value of Re. So I included an interaction term between Re and Fr in our models to account for this relationship.

The third thing I noticed is that weak gravity has a similar effect to no gravity. I see that the lines for Fr = 0.3 and Fr = ∞ are almost identical. So I replaced Fr with a simpler gravity variable that has two categories. In this case, strong gravity (Fr = 0.052), and weak or no gravity (Fr = 0.3 or ∞). I find that this simpler approach predicts just as well as using all three Fr values while making the models easier to understand and interpret.

Models I tried

Model 1: the simplest model, linear in St, plus Re and gravity.

Model 2: the same as model 1 but I used log(St).

Model 3: used a cubic polynomial in log(St), plus Re and gravity.

Model 4: the same as model 3 plus an Re $\times$ gravity interaction.

Model 5: the same as model 4 plus an St $\times$ gravity interaction.

Model 6: the most flexible model, fits a separate St curve for each of the 6 Re-and-gravity combinations.

Model 7 & 8: lasso and ridge versions of model 6, which shrink its coefficients to keep it from overfitting.

Model 1 to model 6, I used ordinary least squares regression to fit these models. For the lasso and ridge regression models, I used cross-validation to select the best value of the shrinkage parameter (λ). I fit every model separately for each of the four statistical measures.

Picking the best model

I used 10-fold cross-validation so that I can compare how well each model predicts new data. I first divided the 89 simulation runs into 10 groups. I then trained each model using 9 groups and tested its predictions on the remaining group. I repeated this process 10 times so that every group was used for testing once. I used the root mean squared error (RMSE) on the log scale so that I can measure prediction accuracy. In this case, an RMSE of 0.1 means that our predictions are usually off by about 10%.

Since I also wanted my models to be easy to understand and interpret, I did not automatically choose the model with the lowest cross-validation error. Instead, for each of the four statistical measures, I selected the simplest model that performed nearly as well as the best model. I also tested polynomial degrees from 1 to 6 to determine how much curvature was needed to capture how St affects the statistical measures. I then selected the degree that produced the lowest cross-validation error.

Final Model

My final model that I ended up selecting are model 4 and model 5. The equation I show you here is model 4. Let $x = \log(St)$. Let $R_{224} = 1$ if $Re = 224$ and $R_{398} = 1$ if $Re = 398$ (both would be 0 if Re = 90). And let $G = 1$ for strong gravity while $G = 0$ is for weak or no gravity.

$$\log(\text{summary}) = \underbrace{b_0 + b_1 x + b_2 x^2 + \cdots + b_d x^d}_{\text{curved effect of St}} + \underbrace{b_{224} R_{224} + b_{398} R_{398}}_{\text{effect of Re}} + \underbrace{b_G G}_{\text{effect of gravity}} + \underbrace{b_{224,G}\, R_{224} G + b_{398,G}\, R_{398} G}_{\text{Re}\times\text{gravity}} + \varepsilon$$

Here, $b_0 + b_1 x + \cdots + b_d x^d$ is the curve that shows how the summary changes with St. With b_{224} and b_{398}, it shifts the whole curve up or down when Re is 224 or 398 instead of 90. b_G represents how

much strong gravity shifts the curve at $Re = 90$. While $b_{224,G}$ and $b_{398,G}$ adjust the gravity effect at $Re = 224$ and $Re = 398$. This is basically the Re $\times$ gravity interaction where gravity matters more at some Re levels than others.

For the mean, the best model is actually model 5, which is the same equation as above plus an St $\times$ gravity interaction.

With d, it shows how curvy the St curve can be and I picked it by trying $d = 1$ to $d = 6$ and kept the one with the lowest CV error. The mean uses model 5 with $d = 4$, the variance uses model 4 with $d = 4$, and I used model 4 for skewness and kurtosis with $d = 2$.

Results

Which model predicts the best?

	Mean	Variance	Skewness	Kurtosis
M1: linear	0.168	1.902	0.608	1.176
M2: log(St)	0.139	1.735	0.614	1.187
M3: cubic in log(St)	0.139	1.688	0.619	1.200
M4: + Re x gravity	0.101	0.791	0.229	0.410
M5: + St x gravity	0.074	0.788	0.277	0.511
M6: separate curves	0.148	0.684	0.231	0.405
M7: lasso	0.091	0.939	0.239	0.411
M8: ridge	0.377	0.971	0.465	0.893
Final model	0.074	0.778	0.219	0.391

d	Mean	Variance	Skewness	Kurtosis
1	0.098	0.972	0.235	0.408
2	0.095	0.832	0.219	0.391
3	0.074	0.791	0.229	0.410
4	0.074	0.778	0.231	0.414
5	0.075	0.786	0.231	0.415
6	0.076	0.788	0.232	0.415

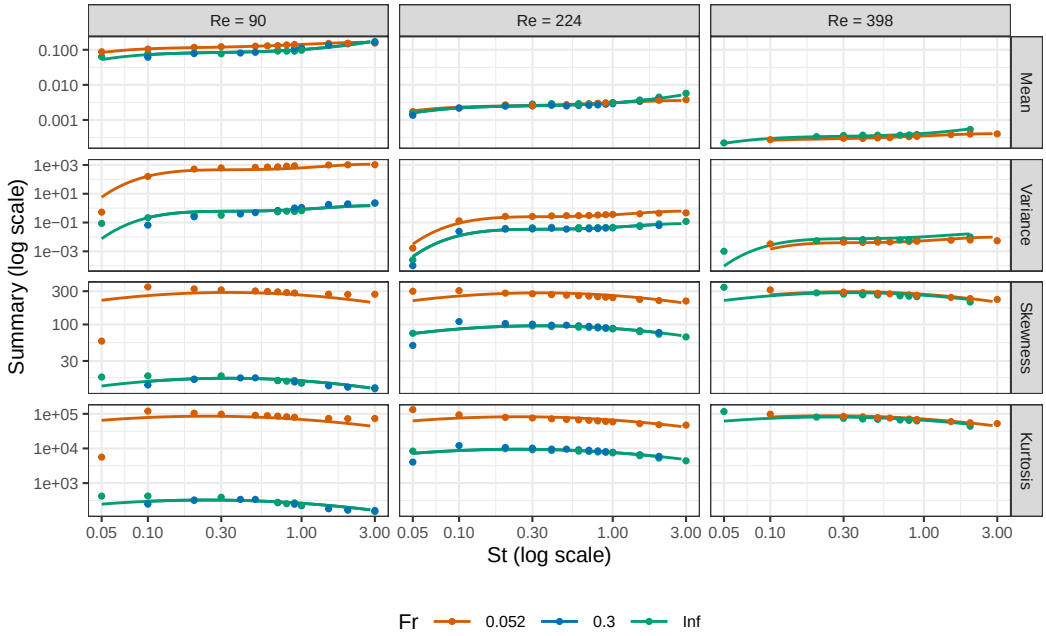
The first table shows the cross-validation error for each model. I see that using $\log(St)$ and adding a curved relationship with St in model 2 and model 3 slightly improves the prediction accuracy. But I do recognized a big improvement coming from model 4, which includes the interaction between Re and gravity. This does reduce the CV error by more than half for variance (1.69 to 0.79), skewness (0.62 to 0.23), and kurtosis (1.20 to 0.41). For the mean, I see that model 5 performs even better by allowing the effect of St to change depending on gravity, which reduced the CV error from 0.101 to 0.074.

With these results, I need to consider the bias-variance tradeoff. With models 1 through 3, the models are too simple to capture important patterns in the data, which would lead to high bias. On the other hand, model 6 is too complex and would begin fitting random noise rather than the actual patterns in the data, which would result in high variance, and this could explain why the CV error for the mean increased.

In the first table, it show models 1 through 6 using a third-degree polynomial of $\log(St)$. So after selecting model 5 for the mean and model 4 for the other variables, I tested the polynomial degrees from 1 to 6 to find which degree gives the most accurate predictions as seen in the second table. The results in the second table showcase that degree 4 performs the best for the mean and variance while degree 2 performs the best for skewness and kurtosis. These choices were used to give me the final model as can be seen in the final row of the first table.

As for uncertainty, the last row of the first table gives the CV error of my final models. Since I model on the log scale, it means that our predictions are typically off by a factor of $e^{\text{CV error}}$. With our final model, the mean has a CV error of 0.074 and $e^{0.074} \approx 1.08$, meaning that my predictions are typically off by about 8%. With this in mind, the prediction errors are approximately 25% for skewness, 50% for kurtosis, and a factor of 2 for variance. But I do note that my predictions are less reliable for test points where St falls outside the range of values used to build the models for their corresponding Re and Fr settings.

How Re, gravity, and St affects the clusters



	Mean	Variance	Skewness	Kurtosis
Re 90 to 224	0.0302	0.0563	5.63	29.3
Re 224 to 398	0.135	0.221	2.98	8.53
Strong vs. no gravity, Re 90	1.43	765	16.9	264
Strong vs. no gravity, Re 224	1.04	7.42	2.96	8.68
Strong vs. no gravity, Re 398	0.866	0.525	1.03	1.09

The plots here shows my models' predictions across St, while the table showcases how much each statistical measures changes when I change one input at a time.

My first big takeaway is that Re has the biggest effect. When Re increases from 90 to 224, I see that the average becomes about 33 times smaller while the skewness increases by about 5.6 times, which suggests that stronger turbulence do lead to more smaller clusters, but there are still a few very large clusters that make the distribution less balanced. Another thing I noticed is that gravity does have a greater effect whenever the turbulence is weaker. As shown from the table, when $Re = 90$, strong gravity increases the variance by about 765 times. But when $Re = 224$, the increase is only about 7 times, and I see that gravity has almost no effect when $Re = 398$. What this could tell us is that gravity causes particles to settle and form larger clusters while stronger turbulence breaks these clusters apart. Another observation to note is that the relationship with St is not linear. And the last thing I want to point out is that the four statistical measures respond differently to the input variable. In the plot, I see that increasing Re does decrease the mean cluster volume, but it also increase skewness and kurtosis. With the graph, I can also tell that gravity has little effect in the mean but significantly affects variance, skewness, and kurtosis. What this tells me is that gravity mainly changes how much the cluster sizes differ instead of significantly affecting their average cluster volume.

Conclusion

In this case study, I used simple linear regression models on the log scale in order to predict the mean, variance, skewness, and kurtosis of particle cluster volumes based on Re, Fr, and St. I compared several models using cross-validation to see which ones made the most accurate predictions. With that, I found that model 4 performs the best for variance, skewness, and kurtosis, which includes a curved relationship with St and an interaction between Re and gravity. For the mean, I found that model 5 performs better since the model accounts for how the relationship between St and the mean cluster volume changes with the strength of gravity. I do find that these models provides accurate predictions while staying it simple enough to understand how each input variable affects the cluster volumes. However, one limitation is that some test points do have St values that are outside the range I used to train the models, which makes the predictions for those points less reliable.

My main finding is that Re has the strongest effect on cluster volumes. As turbulence increases, the average cluster volume is expected to become much smaller, but there are still a few very large clusters that makes the distribution less balanced due to the fact that the skewness and kurtosis increased. Another finding is that gravity has a noticeable effect when it is strong as its influence decreases as turbulence increases. Also, the relationship between St and all four statistical measures is nonlinear, meaning that the effects of St do not follow a linear pattern basically. And the four statistical measures respond differently to the input variables, with gravity having a greater effect on how much cluster sizes varies instead of significantly affecting their average cluster volume.